

How scoring works

Cluster centroids embeds the full corpus, makes n clusters (10/20/50/100), and keeps the centroid post of each cluster so you get n posts.

Label coverage labels every post (mental status, emotion, size) and picks 10/20/50/100 posts that cover those label combinations.

Vendi then scores the two selected sets in the same embedding space. Higher means the sample is more diverse.

Size bands: small 10–50, medium 51–200, large 201–500 words (selftext). Outside that range is out_of_band.

Recent runs

id	created_at	subredd
13	2026-09-20T23:43:30	Opiatesl
12	2026-09-20T23:38:38	Opiatesl
11	2026-09-20T23:38:14	Opiatesl
10	2026-09-20T23:30:45	Opiatesl
9	2026-09-20T23:30:15	Opiatesl
8	2026-09-20T23:25:01	Opiatesl
7	2026-09-20T23:24:38	Opiatesl
6	2026-09-20T23:23:39	Opiatesl

Reddit Post Diversity Compare

First scrape a subreddit and save a raw CSV, then sample with cluster centroids or label coverage and compare them with Vendi.

1. Scrape posts

Subreddit

OpiatesRecovery

Listing

Hot

Max listing posts to fetch

1000

Batch size

100

Pause between batches (seconds)

60

☐ Include stickied posts

☒ Only keep posts with selftext

Comment sort

top

Comments to save per post

3

Saved raw files for this subreddit

OpiatesRecovery_193_raw.csv

☐ Re-scrape even if a raw file already exists

Scrape and save

Loaded 193 posts from OpiatesRecovery_193_raw.csv.

Raw file naming

Files are saved under `data/` as `subreddit_postcount_raw.csv`. Example: `OpiatesRecovery_847_raw.csv`.

Scrape options

- Fetch in batches, then pause so Reddit rate limits are respected
- Skip stickied / empty posts unless you turn those filters off
- Comments are optional; 0 skips comment requests

Embeddings (`sentence-transformers/all-roberta-large-v1`), mental status (`SajjadIslam/multiMentalRoBERTA-6-class`), and emotion (`tasinhoque/roberta-large-go-emotions`) run later, when you sample. Credentials come from the repo `.env`.

Already in `data/` : `OpiatesRecovery_193_raw.csv`, `OpiatesRecovery_94_raw.csv`

Posts collected

File name

Folder

193

OpiatesRecovery_...

data/

Full path: data/OpiatesRecovery_193_raw.csv · r/OpiatesRecovery · hot

Raw scrape preview

created	title	author	score	words	size	post
2026-09-19 17:00 UTC	When do you actually know you are back at 100%?	Ok_Date6167	9	121	medium	https://www.reddit.com
2026-09-19 21:03 UTC	Scared to Ask, But...	Bombsaway00	1	124	medium	https://www.reddit.com
2026-09-19 10:15 UTC	Day 7 WD	Realistic-Bit-4933	5	594	out_of_band	https://www.reddit.com
2026-09-19 03:55 UTC	I just found y'all today	AlwaysDeep469	10	152	medium	https://www.reddit.com
2026-09-18 18:27 UTC	30 days clean - Bought 10 pills, took 2, flushed the re	TahoeTweezer4EVA	13	193	medium	https://www.reddit.com
2026-09-18 20:53 UTC	Five whole days	MaybeLeast	5	72	medium	https://www.reddit.com
2026-09-18 18:54 UTC	Day three Fentanyl patch withdrawal.	CannaBli55	6	293	large	https://www.reddit.com
2026-09-18 19:25 UTC	Friday September 18 check in	xzxnightshade	4	131	medium	https://www.reddit.com
2026-09-18 11:30 UTC	Day 6 WD	Realistic-Bit-4933	4	332	large	https://www.reddit.com
2026-09-18 05:05 UTC	What actually helped	Prudent_Yam_7667	3	406	large	https://www.reddit.com

Download raw CSV

2. Run Cluster centroids or Label coverage

Sample size / Cluster centroids clusters (k) ?

☒ 10 ☐ 20 ☐ 50 ☐ 100

ChromaDB already has 193 / 193 embeddings for r/OpiatesRecovery. No new embedding work is needed.

Models will use GPU: NVIDIA RTX 2000 Ada Generation

Run Cluster centroids

Run Label coverage

✓ Label coverage Vendi 6.91

Cluster centroids

Created 10 clusters and kept 10 centroid posts (the post nearest each cluster center). Vendi: 5.80

Cluster centroids Vendi 

5.80

Cosine Vendi · 10 centroid posts · 58% of maximum (10)

r/OpiatesRecovery · 10 centroid posts · mean cosine 0.391 · embeddings from ChromaDB ([data/chroma](#))

created	title	author	score	words	size	cluster	selection
2026-09-16 12:50 UTC	WD morning 4	Realistic-Bit-4933	15	125	medium	cluster_1	cluster_1
2026-08-15 17:27 UTC	Sat/Sun August 15/16 check in	xzxnightshade	1	139	medium	cluster_2	cluster_2
2026-08-30 01:13 UTC	I finally shook my oxy addiction after a year of trying	jlaay	34	488	large	cluster_3	cluster_3
2026-08-21 17:34 UTC	Friday August 21 check in	xzxnightshade	2	96	medium	cluster_4	cluster_4
2026-09-11 10:53 UTC	Day 11-12	Nervous_Complaint_30	3	52	medium	cluster_5	cluster_5
2026-09-14 13:28 UTC	I'm devastated in myself.	Realistic-Bit-4933	3	136	medium	cluster_6	cluster_6
2026-09-15 20:27 UTC	90 days clean	Comfortable_Day737	10	114	medium	cluster_7	cluster_7
2026-09-13 20:08 UTC	Day 13	Nervous_Complaint_30	6	137	medium	cluster_8	cluster_8
2026-09-12 01:21 UTC	Want to take control over my life again.	Ambitious_Farm_1810	4	306	large	cluster_9	cluster_9
2026-08-10 20:49 UTC	I hope someone hears me on this! What is the difference	Content_Bus6004	3	181	medium	cluster_10	cluster_10

Posts per cluster in the corpus (one centroid kept from each)

cluster	post_count
cluster_1	17
cluster_2	5
cluster_3	37
cluster_4	35
cluster_5	14
cluster_6	20
cluster_7	13
cluster_8	19
cluster_9	26
cluster_10	7

Cosine similarity of cluster centers

	cluster_1	cluster_2	cluster_3	cluster_4	cluster_5	cluster_6	cluster_7	cluster_8	cluster_9	cluster_10
cluster_1	1	0.195	0.622	0.446	0.69	0.705	0.614	0.732	0.694	0.374
cluster_2	0.195	1	0.17	0.434	0.149	0.141	0.26	0.143	0.085	0.116
cluster_3	0.622	0.17	1	0.21	0.687	0.822	0.728	0.809	0.812	0.629
cluster_4	0.446	0.434	0.21	1	0.297	0.21	0.451	0.293	0.248	0.134
cluster_5	0.69	0.149	0.687	0.297	1	0.782	0.638	0.811	0.825	0.428
cluster_6	0.705	0.141	0.822	0.21	0.782	1	0.655	0.807	0.815	0.51
cluster_7	0.614	0.26	0.728	0.451	0.638	0.655	1	0.752	0.651	0.529
cluster_8	0.732	0.143	0.809	0.293	0.811	0.807	0.752	1	0.811	0.559
cluster_9	0.694	0.085	0.812	0.248	0.825	0.815	0.651	0.811	1	0.454
cluster_10	0.374	0.116	0.629	0.134	0.428	0.51	0.529	0.559	0.454	1

Save 10 posts to data/OpiatesRecovery_10_a1.csv

Download Cluster centroids CSV

File: data/OpiatesRecovery_10_a1.csv

Label coverage

Vendi score: 6.91 · 10 posts across label groups

Label coverage Vendi 

6.91

Cosine Vendi on size × mental × emotion coverage · 69% of maximum (10)

r/OpiatesRecovery · 10 selected · 74 label groups · mean cosine 0.275

created	title	author	score	words	size	mental status
2026-09-19 10:15 UTC	Day 7 WD	Realistic-Bit-4933	5	594	out_of_band	depression
2026-08-19 19:54 UTC	A fight that startet 10 years ago	Halexorus	4	551	out_of_band	depression
2026-09-15 12:20 UTC	Morning 3 of yet another WD just ranting	Realistic-Bit-4933	9	131	medium	depression
2026-09-10 14:33 UTC	Day 10	Nervous_Complaint_30	3	80	medium	depression
2026-09-13 20:06 UTC	Seeking Anxiety Tips	PLURdude1984	1	62	medium	anxiety
2026-08-09 06:44 UTC	Does saffron actually work after and during withdrawals ?	Excellent-Walrus5781	1	3	out_of_band	none
2026-08-21 15:48 UTC	How long are withdrawals after six weeks of post op use	ladylokaaa	5	33	small	ptsd
2026-09-11 06:36 UTC	Help!!	saturn1ascends	3	376	large	depression
2026-09-10 17:46 UTC	Uk based painkiller addict - suggestions on how to be free	skipster88	3	356	large	depression
2026-08-27 07:38 UTC	What's your take on adrenaline junkies?	super-renata	5	186	medium	depression

Post size

Emotion

size	posts	emotion	posts
medium		surprise	2
out_of_band		annoyance	1
large		neutral	1
small		optimism	1
Mental status		sadness	1
		gratitude	1
		curiosity	1
		confusion	1
		disappointment	1

mental status	posts
Download Label coverage CSV	7
anxiety	1
Compare	1
...	1

CLUSTER CENTROIDS

One post nearest each of k clusters

5.80

Cosine Vendi · k=10 clusters · 58% of maximum (10)

LABEL COVERAGE

Size × mental status × emotion coverage

6.91

Cosine Vendi on the selected set · 69% of maximum (10)